

Problem

If $A = 3 - j4$, find: (a) $A^{1/3}$ (3 roots), and (b) $\ln A$.

Step-by-step solution

Step 1 of 3

Consider the complex number, $A = 3 - j4$.

To calculate $A^{1/3}$, convert the complex number A into polar form.

Write the generalized polar form representation of complex number, $z = x + jy$.

$$z = r \angle \theta \quad \left(r = \sqrt{x^2 + y^2}, \theta = \tan^{-1} \left(\frac{y}{x} \right) \right)$$

Calculate the magnitude of complex number A .

$$\begin{aligned} A &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Calculate the angle θ_1 , as the complex number $A = 3 - j4$ fall in fourth quadrant.

$$\begin{aligned} \theta_1 &= 360^\circ - \tan^{-1} \left(\frac{4}{3} \right) \\ &= 360^\circ - 53.15^\circ \\ &= 306.84^\circ \end{aligned}$$

Hence the polar form of a complex number is $A = 5 \angle 306.84^\circ$.

Step 2 of 3

(a)

Calculate the value of $A^{1/3}$, which has three roots.

$$\begin{aligned} A^{1/3} &= 5^{1/3} \angle \left(\frac{306.84^\circ}{3} + \frac{2\pi k}{3} \right) \quad (\text{here } k = 0, 1, 2) \\ &= 1.71 \angle (102.28^\circ + 0), 1.71 \angle \left(102.28^\circ + \frac{2\pi}{3} \right), 1.71 \angle \left(102.28^\circ + \frac{2\pi(2)}{3} \right) \\ &= 1.71 \angle 102.28^\circ, 1.71 \angle (102.28^\circ + 120^\circ), 1.71 \angle (102.28^\circ + 240^\circ) \\ &= 1.71 \angle 102.28^\circ, 1.71 \angle (222.28^\circ), 1.71 \angle (342.28^\circ) \end{aligned}$$

Hence, the three roots of $3 - j4$ is $\boxed{1.71 \angle 102.28^\circ, 1.71 \angle 222.28^\circ, 1.71 \angle 342.28^\circ}$

Step 3 of 3

(b)

Write the general solution of $\ln(re^{j\theta})$.

$$\ln(re^{j\theta}) = \ln r + j\theta + j2k\pi \quad (\text{here } k \text{ as an integer})$$

Calculate the value of $\ln A$.

$$\begin{aligned}\ln(A) &= \ln(5e^{j306.84^\circ}) \\ &= \ln(5) + \ln(e^{j306.84^\circ}) \\ &= 1.609 + j306.84^\circ + j2k\pi\end{aligned}$$

Convert the angle in degrees into radians.

$$\begin{aligned}306.84^\circ &= 306.84^\circ \left(\frac{\pi}{180^\circ} \right) \\ &= 1.7047(3.14) \\ &= 5.356^\circ\end{aligned}$$

Thus, the value of $\ln A$ is $\boxed{1.609 + j5.356 + j2n\pi \ (n = 0, 1, 2, 3, \dots)}$.